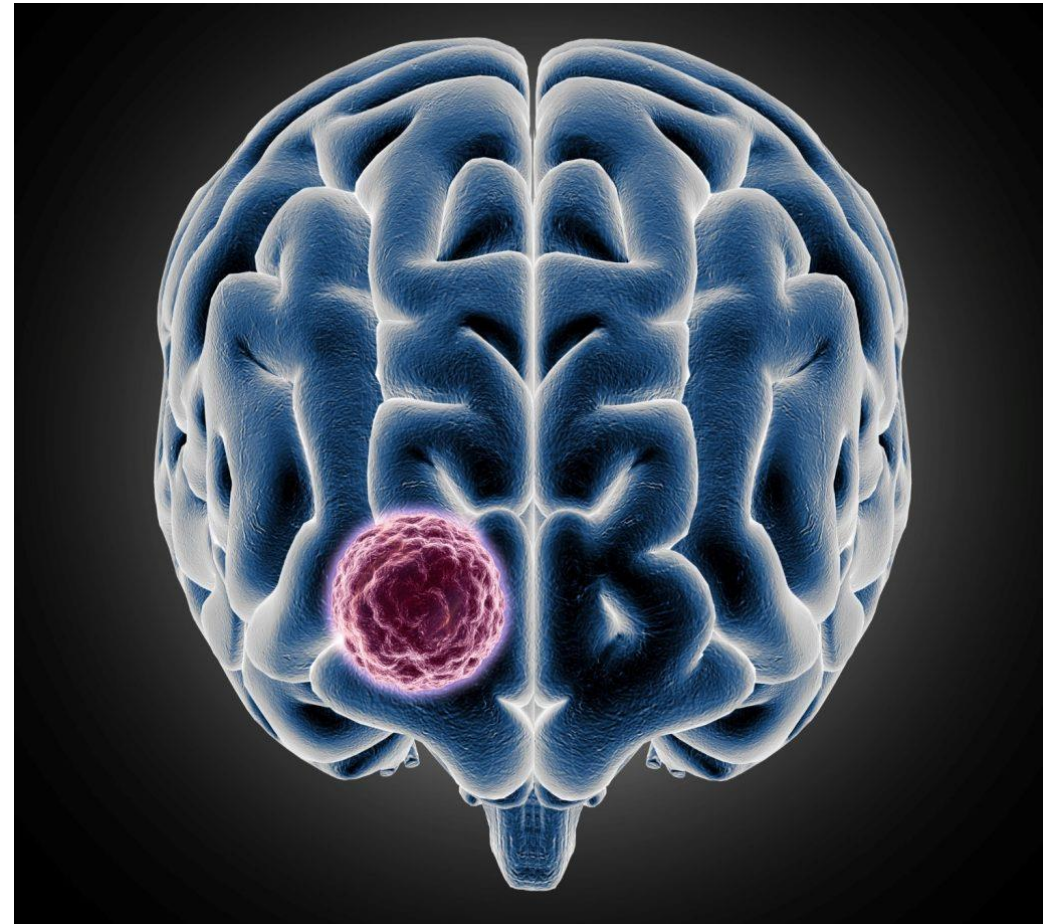


“Reliable and Explainable AI for Brain Tumor Detection Using Graph Neural Networks”



Presented by: **Neha Mahanand**

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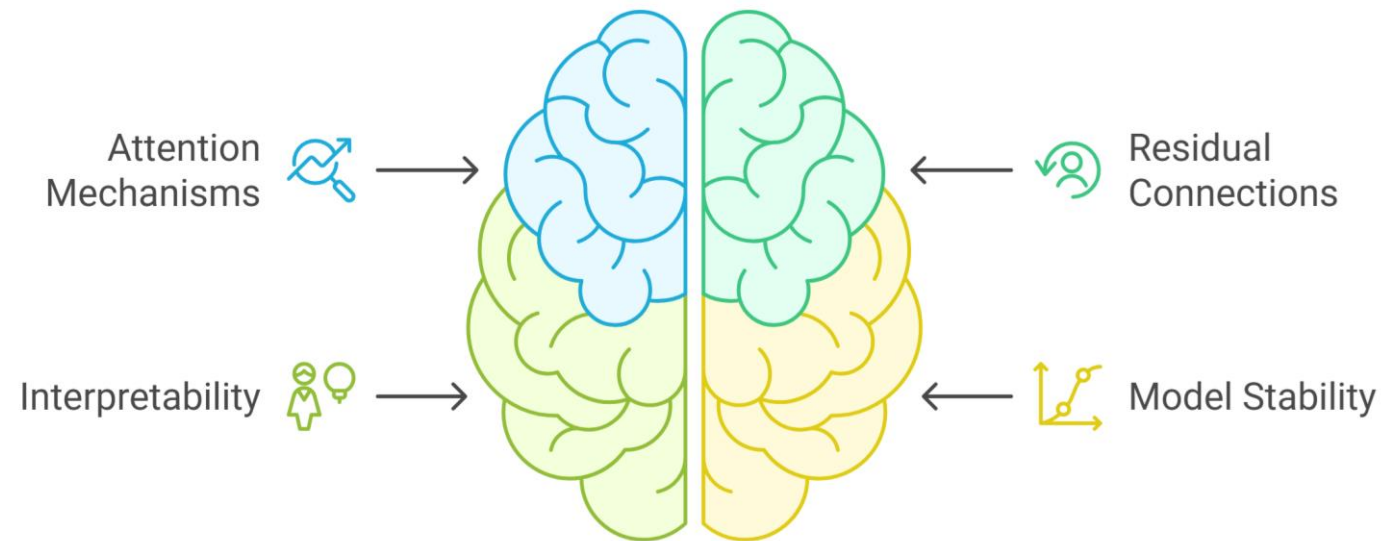
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1. Project Background

1.1 Introduction & Objective

- While conventional CNNs give high accuracy but poor interpretability, Explainable AI (XAI) adds transparency and trust to AI-based medical diagnosis.
- This work aims to develop a GATv2-based Graph Neural Network for accurate brain tumor classification, using attention mechanisms and residual connections for better interpretability and model stability.



2. Methods and Work Flow

2.1 Dataset

- The dataset is from Kaggle
- Number of class: 4
- ['glioma_tumor', 'meningioma_tumor', 'no_tumor', 'pituitary_tumor']
- These are encoded using label encoding

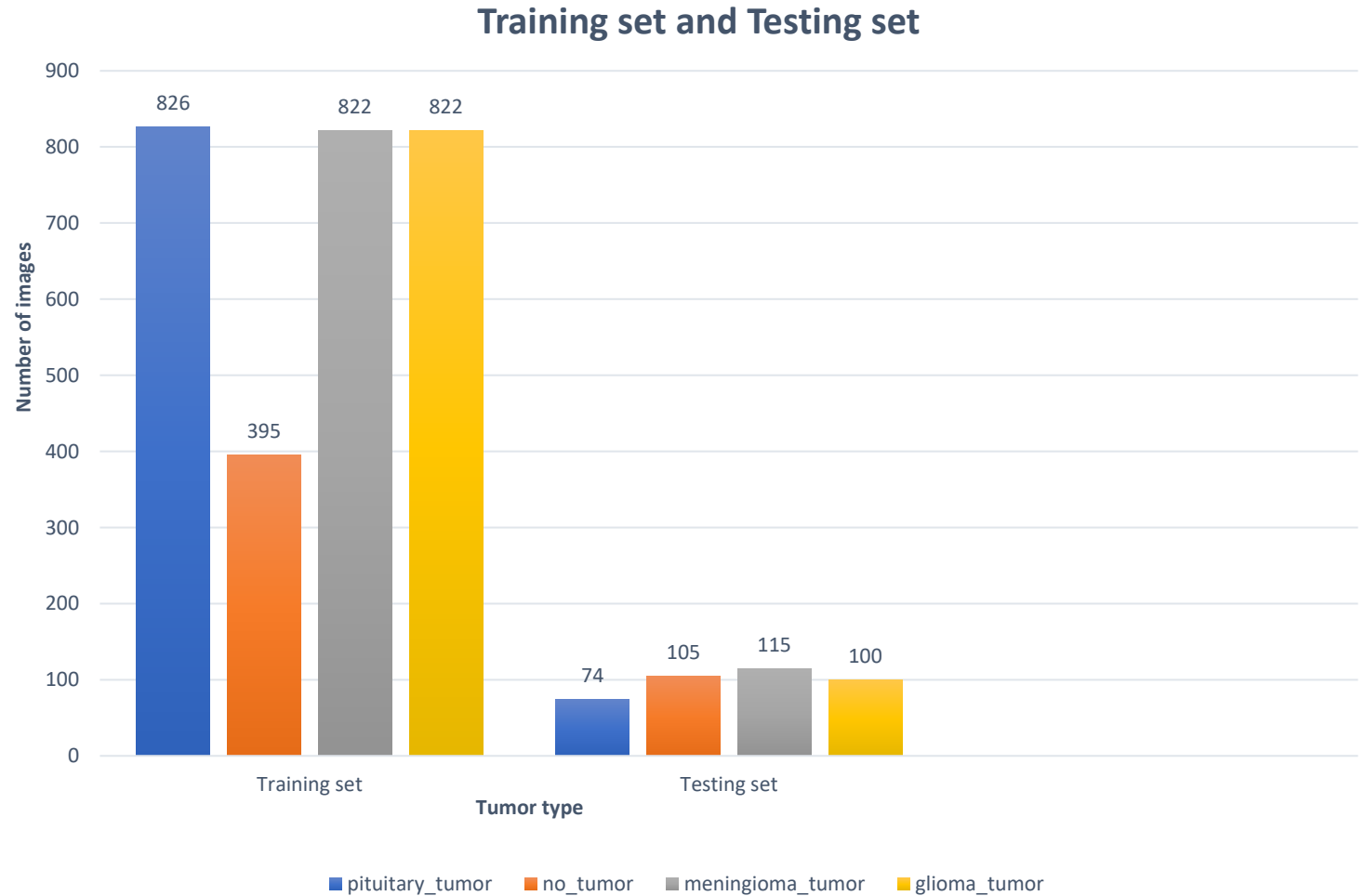
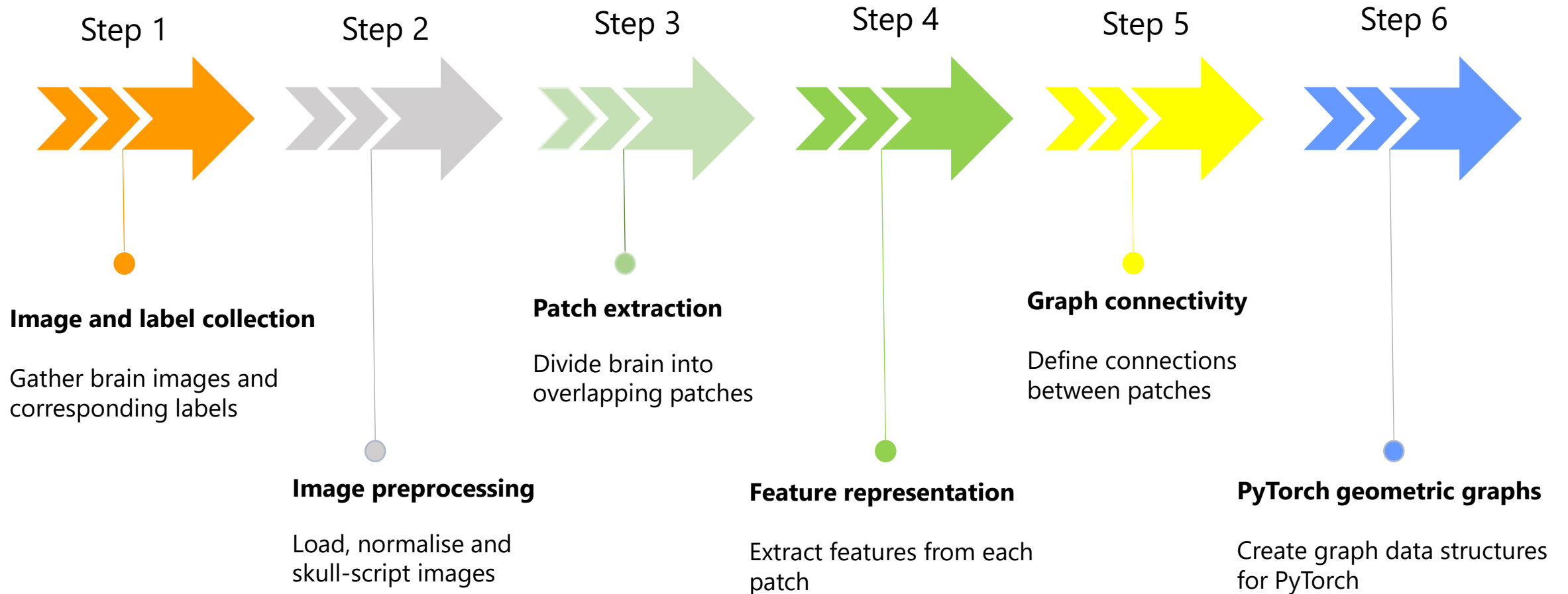


Figure 1: Training set and testing set

2.2 Image processing & Graph construction



2.3 Image conversion pipeline

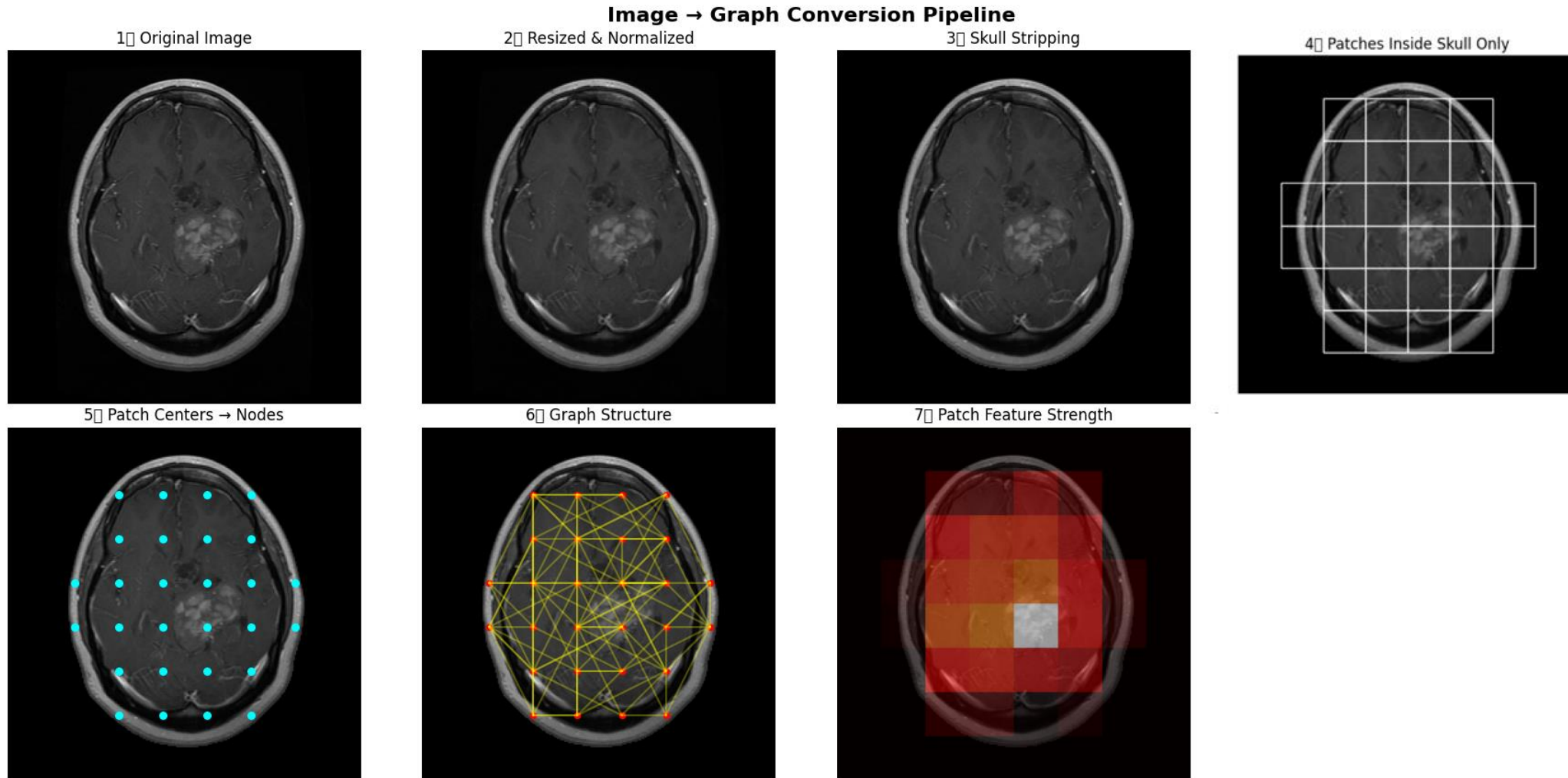


Figure 2: Example of Image to graph conversion pipeline

2.4 Model Overview and Architecture

- Graph Attention Network v2 (GATv2)
- Brain MRI images are divided into patches, which are treated as nodes in a graph.
- Edges connect neighbouring patches, allowing the model to capture spatial relationships in the brain.
- GATv2 layers learn relationships between patches using an attention mechanism.
- Attention weights allow the model to focus more on important brain regions relevant for tumor classification.

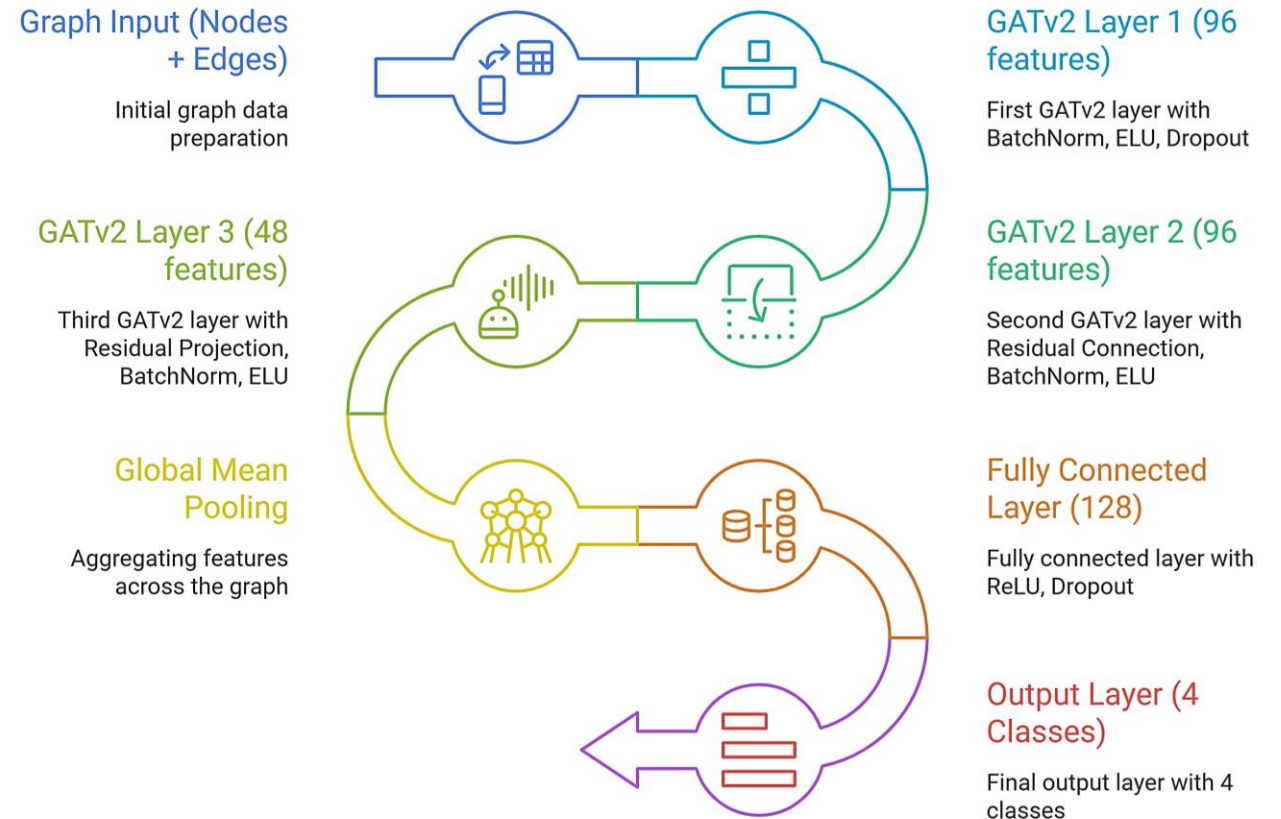
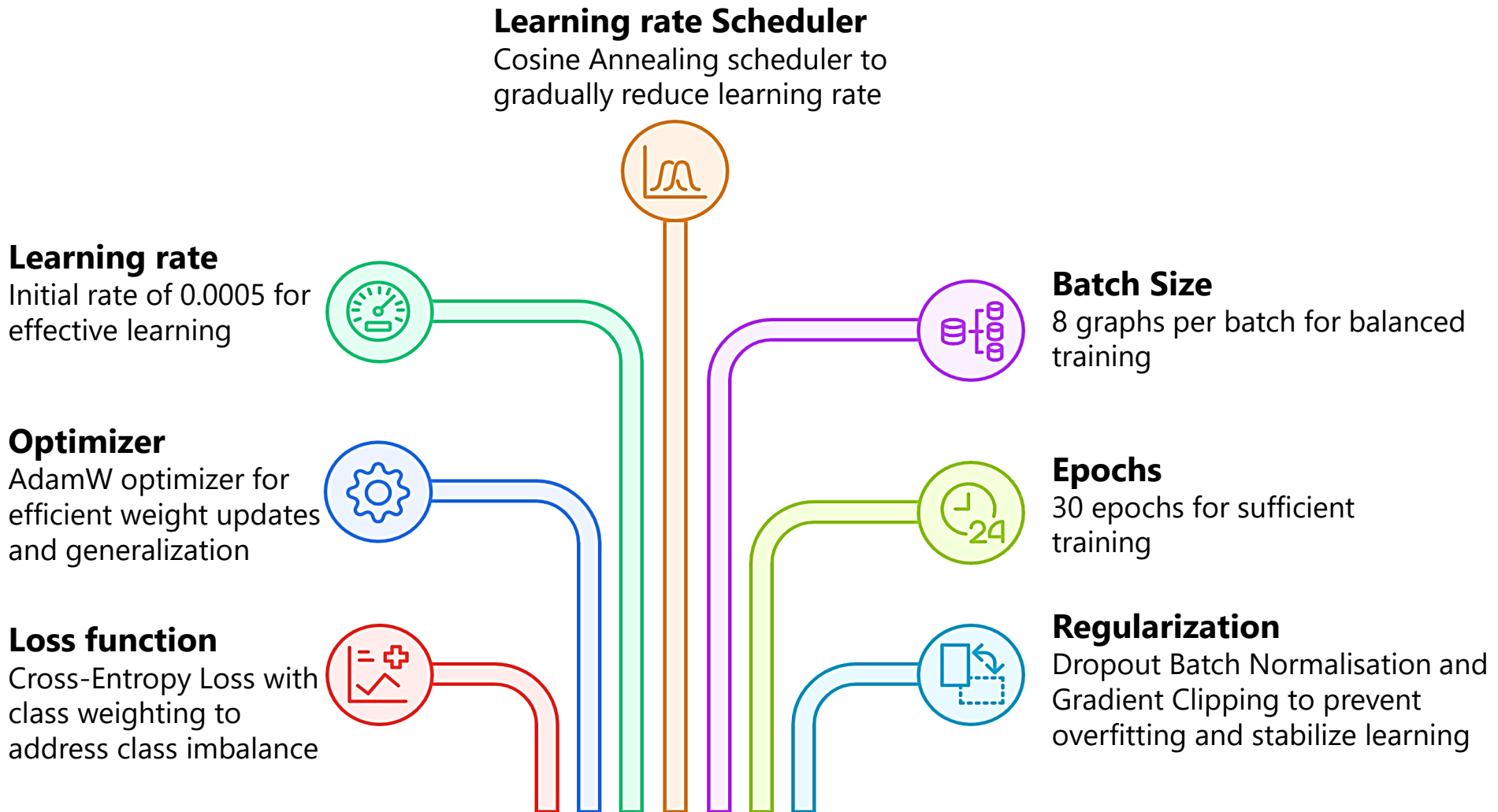


Figure 5 : Model Architecture

2.5 Model Training Strategy



2.6 Evaluation

Classification Report:

	precision	recall	f1-score	support
glioma_tumor	0.33	0.13	0.19	100
meningioma_tumor	0.44	0.55	0.49	115
no_tumor	0.40	0.70	0.51	105
pituitary_tumor	0.61	0.23	0.33	74
accuracy			0.42	394
macro avg	0.44	0.40	0.38	394
weighted avg	0.43	0.42	0.39	394

Overall Accuracy: 0.4213

Figure 4 : Classification report

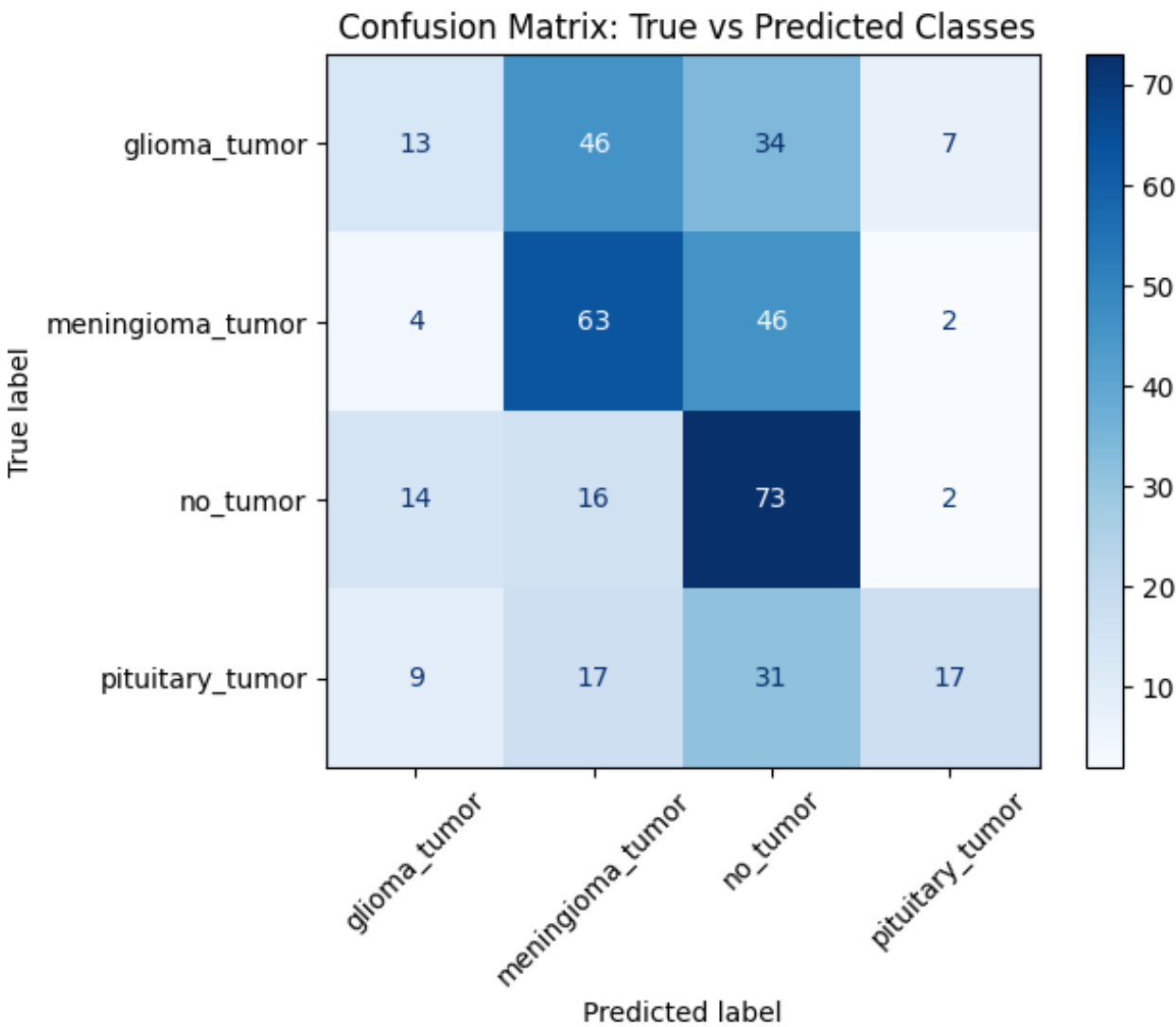


Figure 6 : Confusion matrix

2.7 Grad-CAM Heatmap for Graph Node Importance

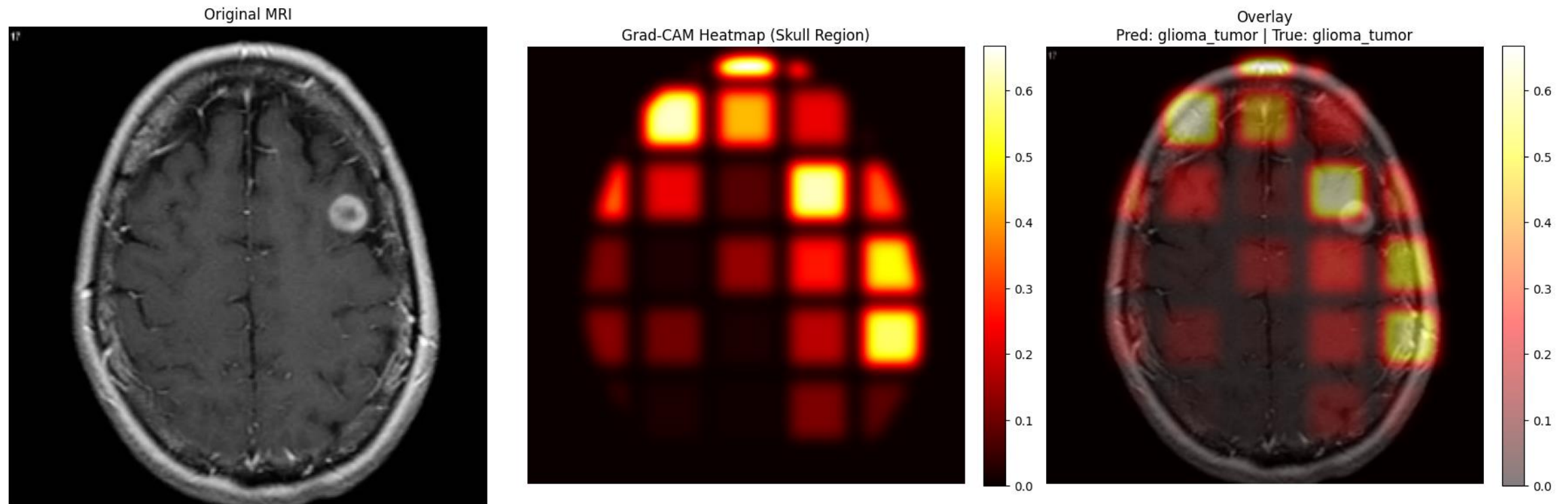


Figure 7 : Example of test image

2.8 GAT v2 Node Importance Visualization

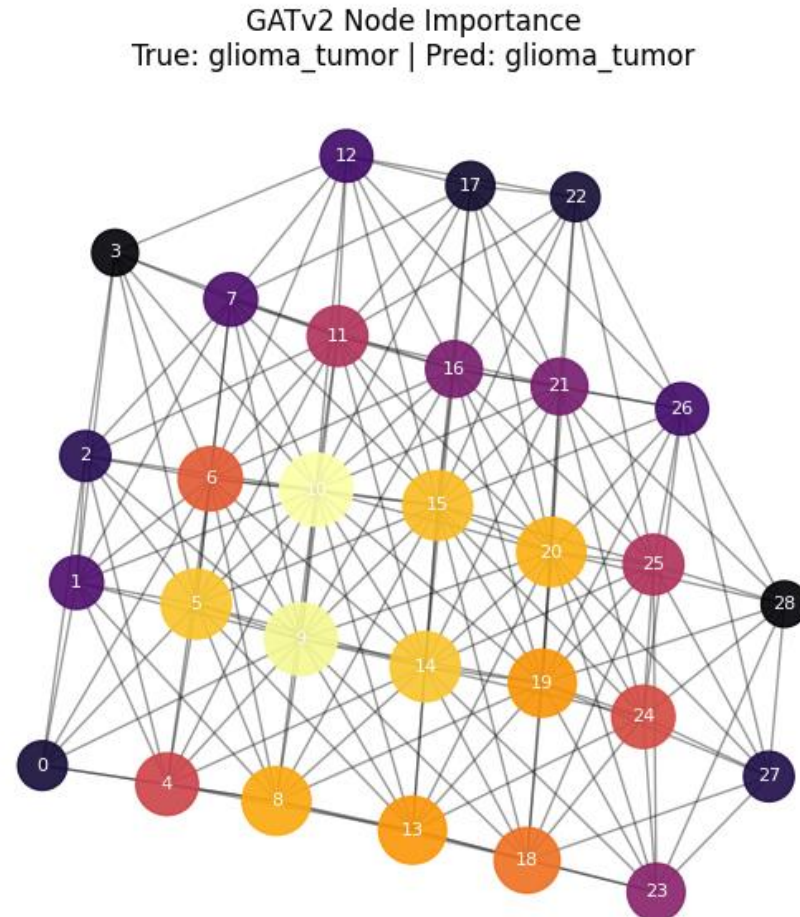


Figure 8 : Example of test image

- Nodes represent MRI patches / features
- Edges shows the relationship between patches
- Import nodes are represented in light colours
- Position of nodes are random layout, actual MRI location is not preserved

3.Results & Discussions

- Overall model accuracy: **42.13%**
- Better performance in **no-tumor detection** compared to tumor classification
- **Glioma tumor** detection is **weak** (recall = 0.13)
- MRI tumors share **similar visual patterns**, making classification challenging
- **Limited dataset size** affects model generalization
- With further improvements, such as:
 - enhanced feature extraction,
 - additional training data,
 - improved model architecture,
 - advanced techniques like transfer learning and
 - data augmentationcan be implemented to improve classification accuracy and tumor detection capability.

4. References

- Dataset Kaggle link (<https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri>)
- V. Ponzi and C. Napoli, "[Graph Neural Networks: Architectures, Applications, and Future Directions](#)," in IEEE Access, vol. 13, pp. 62870-62891, 2025, doi: 10.1109/ACCESS.2025.3558752.
- Ali, S., Abuhmed, T., El-Sappagh, S., Muhammad, K., Alonso-Moral, J. M., Confalonieri, R., Guidotti, R., Del Ser, J., Díaz-Rodríguez, N., & Herrera, F. (Year). [Explainable Artificial Intelligence \(XAI\): What we know and what is left to attain Trustworthy Artificial Intelligence](#). Journal/Publisher. DOI <https://doi.org/10.1016/j.inffus.2023.101805>

Thank you